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Frege on the Statement of Number

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Recent interest in the history of analytic philosophy has focused attention quite naturally on the "founding fathers" of analysis: Moore, Russell, Wittgenstein, and Carnap. This, in turn, has led to a re-examination of the nineteenth century and to intense speculation about the work of the German mathematician Gottlob Frege (1848-1925). Yet Frege's proper position within the historical canon of the discipline has been difficult to determine. If Frege is a philosopher, why are philosophical doctrines, particularly traditional ones, so hard to find in Frege's writings — even the *Grundlagen*?

Criticism too has surrounded the very prospect of a historical account of Frege's work. Some have questioned whether a credible investigation can be made into Frege's intellectual background or whether such an investigation, once made, can be adequately evaluated. These sceptics fear that the only connections historical research will be able to bring to light are "parallels," that is, merely analogical features. What such an investigation will be unable to determine are definitive "sources" for any of the key ideas in Frege's thought. Instead, it will be limited to making comparisons, possibly superficial, between the ideas of Frege and those of his predecessors (or successors).

The search for distinct sources quickly fixes upon the few direct references to works and figures that Frege actually proffers. This brief paper will do likewise in its examination of a possible source for Frege's central doctrine that "the content of a statement of number is an assertion about a concept."[1] This doctrine provides a key element in Frege's logicist project: it is, as Frege himself later described it, the "most fundamental" of his results on the concept of number and that upon which his account in the *Grundgesetze* "rests."[2] But could this fundamental idea conceivably have appeared *before* Frege?

[1] Gottlob Frege, *The Foundations of Arithmetic*, trans. J. L. Austin (Oxford: Basil Blackwell, 1974), p. 59.

[2] Gottlob Frege, *The Basic Laws of Arithmetic*, ed. and trans. Montgomery Furth (Berkeley and Los Angeles: University of California Press, 1967), p. 5.

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It did. In one of his major works, the *Psychologie als Wissenschaft*, the now obscure nineteenth-century philosopher Johann Friedrich Herbart (1776-1841) set forth a theory of number surprisingly similar to Frege's, in direct opposition to the one suggested by Kant. Kant, remember, had characterized number as the result of our conscious awareness of the successive addition of homogeneous units. For Kant, number, generally speaking, is a "plurality considered as unity."[3] It is a unity which, in counting, is dependent upon the generation of time.[4]

Herbart is unsympathetic to this viewpoint or to any account of number which is based upon the idea of combining units in time to form a totality. Herbart starts off his discussion in the *Psychologie* by noting that "since Kant, it has often enough been repeated that the successive addition of one to one yields this idea [of number], which herewith is connected with *time*."[5] But he is driven to conclude his own investigation with the thesis that number "has no more in common with time than do a hundred other types of ideas, which also can be produced only gradually."[6] How then is he able to remove the concept of number from the domain of time?

Herbart effects this separation by divorcing the concept of number proper from that of homogeneous units. According to Herbart, time enters the picture in the "common operation of counting," for in counting it appears that "the number $n + 1$ is produced from the immediately preceeding number n through the addition of the unit."[7] But this picture assumes that numbers actually consist of units. Abandon this assumption, Herbart believes, and one removes the need for temporal succession. To this end, Herbart brings to light an often overlooked feature of our statements of number; this feature is that:

> . . . during counting, something is always present which is being counted; and the idea of this something must always remain similar, since notedly dissimilar things (for example, quills, sheets of paper, sticks of sealing-wax) cannot be counted together unless one regards them as similar (by means of the general concept of 'writing materials').[8]

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[3] Immanuel Kant, *Critique of Pure Reason*, trans. Norman Kemp Smith (New York: St. Martin's Press, 1965), p. 116 [B 111].

[4] Kant, *CPR*, p. 184 [B 182].

[5] Johann Friedrich Herbart, *Sämtliche Werke in chronologischer Reihenfolge*, ed. Karl Kehrbach and Otto Flügel (reprinted Aalen: Scientia Verlag, 1964), vol. 6, p. 110. All translations from German-only sources are the responsibility of this writer.

[6] Herbart, *SW*, vol. 6, p. 111.

[7] Herbart, *SW*, vol. 6, p. 110.

[8] Herbart, *SW*, vol. 6, p. 111.

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That is to say, whenever we count, or make a statement of number, there must be something *that* we count: this something will always be indicated for us by the general concept which appears in those statements.

But then what is counted or collected together are not the elusive units, but rather objects picked out by that general concept. Number, Herbart alleges, needs to rely on the "homogeneous," but not on "homogeneous units." Rather, number is now revealed to have an intimate connection to the general concept. Hence, our numerical predications may be characterized by the fact that in them:

> Each number . . . relates to a general concept of the counted. This concept can, however, remain entirely undetermined, since it is completely irrelevant to the determination of number *what* one is counting. However, one must always keep in mind the relation to this undetermined concept, since one would otherwise be seduced by the false conception of units as components of numbers. To the number 12 one may add in thought the general concept of 'a chair' or 'a thaler' and so one would become aware that the determination of number is undivided and at once attaches to the concept . . . one can by all means successively count through the dozen . . . but the number 12 does not consist of twelve units, since the unit would assume in this way the place of the general concept of the countable.[9]

As should be apparent, this view bears a strong resemblance to the one which Frege adopts as his own: in making a statement of number, Frege claims that we are in fact making an assertion about (a predication of) a concept.[10] So when Herbart concludes the above-cited passage with the remark that "a number is . . . one of the possible answers to the question: How many?" it is no surprise that here too we find a prominent motif of the *Grundlagen*: "What answers the question How many? is number."[11]

Of course, if Frege never read Herbart or was otherwise unaware of his doctrines, then the effect of this parallelism of theme would be nugatory. But Frege *did* read Herbart — and for this claim there are several pieces of evidence. First, among the recorded items of the Nachlass (now destroyed) we find that Frege made excerpts from Herbart specifically on the topic of the "sense of a statement of number."[12] Second, Frege's yet wider knowledge of Herbart may be indicated by his occasional reference to Herbartian concepts, such as the "travelling limit."[13] Third, Frege

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[9] Ibid.

[10] We are in fact ascribing to our general concept a certain property: as Frege explains it, our general concept is subsumed by a "higher or, so to say, second order concept." See Frege, *FA*, p. 65.

[11] Frege, *FA*, p. 57. See also pp. 5n, 26, and 98.

[12] Albert Veraart, "Geschichte des wissenschaftlichen Nachlasses Gottlob Freges und seiner Edition. Mit einem Katalog des ursprüglichen Bestands der nachgelassenen Schriften Freges," *Studien zu Frege I*, ed. Matthias Schirn (Stuttgart: Frommann-Holzboog, 1976), pp. 96 and 103.

[13] Gottlob Frege, *Collected Papers on Mathematics, Logic, and Philosophy*, ed. Brian

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conspicuously announces his allegiance to Herbart's doctrine on the concept of number at the very beginning of the *Grundlagen*.

In the Introduction, Frege worries that few people even consider "the concept of positive whole number" to be problematic, assuming instead that "this concept is adequately dealt with in the elementary textbooks, where the subject is settled once and for all."[14] Frege insists that we turn away from this "schoolboy" mentality and examine "what others have thought." The result of our not doing so is

> . . . that we still rest content with the crudest of views, even though since Herbart's day a better doctrine has been available. It is sad and discouraging to observe how discoveries once made are always threatening to be lost again in this way, and how much work promises to have been done in vain, because we fancy ourselves so well off that we need not bother to assimilate its results.[15]

Now the footnote to Herbart's name, marked in the text, does not refer to the passages cited above from the *Psychologie* but to an educational work, the *Umriss pädagogischer Vorlesungen*.[16] Do the relevant passages from the *Umriss* include statements similar or equivalent to those already disclosed?

They do indeed. The applicable quote — which Frege indicates with the extract, "Two does not mean two things, but doubling" — again sharply disavows any view of numbers as "sums of units" and promulgates the doctrine that numerical predications are composed instead of "the undivided number" and "the general concept":

> Still more might be said against that false view of numbers, as if they were sums of units. This is so as little as that sums are products. Two does not mean two things, but doubling, no matter if the doubled is one or many. The concept of '*a* dozen chairs' does not comprise twelve ideas of *single* chairs; rather there are contained only two ideas: the general concept '*chair*' and the undivided multiplication by twelve. The concept of 'one hundred men' likewise contains only two concepts: the general concept 'man' and the undivided number 'one hundred'. Likewise 'six foot', 'seven pound' — expressions in which language comes to our aid with the use of the singular.[17]

McGuinness and trans. Max Black *et al.* (Oxford: Basil Blackwell, 1984), p. 178. The term derives ultimately from Herbart, *SW*, vol. 8, p. 68.

[14] Frege, *FA*, p. iii.

[15] Ibid.

[16] Frege's possible educational interests — and their root in Herbart's work — would not be unusual for the time. Since 1865, there had been a strong revival of enthusiasm for Herbart's writings on scientific pedagogy. Karl Volkmar Stoy (1815-1885), a Herbartian, founded a pedagogical seminar at the University of Jena in 1874. After Stoy's retirement, Wilhelm Rein (1847-1929), also a Herbartian and one of Frege's colleagues, turned Jena into a Mecca for educators all over the world. On these and other points of interest, see Harold Dunkel, *Herbart and Herbartianism* (Chicago: University of Chicago Press, 1970).

[17] Herbart, *SW*, vol. 10, p. 102.

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Notice that Herbart suggests that the truth of his doctrine may be illustrated by the appearance of the singular in certain linguistic constructions. This thought Frege also strongly echoes when he remarks that corroboration for his central doctrine may be found in German idiom, that the "use of the singular . . . may indicate that the concept is intended, not the thing."[18]

Furthermore, we should also recall that in the *Grundlagen* Frege does not present the central doctrine exclusive of a long section devoted to a discussion of number's relation to the unit. In fact, Frege put forth his doctrine as a "solution" to a very particular "difficulty." This difficulty Frege specifies as that of reconciling the purported identity of units with their distinguishability:

> If we try to produce the number by putting together different distinct objects, the result is an agglomeration in which the objects contained remain still in possession of precisely those properties which serve to distinguish them from one another; and that is not number. But if we try to do it in the other way, by putting together identicals, the result runs perpetually together into one and we never reach a plurality.[19]

We are bound, Frege thinks, to uncover a solution to this difficulty that will preserve the "distinction between one and many."[20]

> Our dilemma is that if we give the name 'unit' to the things to be numbered, then, strictly speaking, we cannot call them identical; and yet if they are identical, then we can never get beyond the number one. That is to say, if the things to be numbered have distinguishing properties, then they cannot be, by definition, the same thing and yet if they have no distinguishing properties, then there is no way to reach a plurality. In the attempt to fix upon homogeneous units either we "obliterate" the distinction between the one and the many we were bound to preserve, or we are "forced . . . to ascribe to units two contradictory qualities, namely identity and distinguishability."[21]

The solution Frege prescribes, which relies upon the central doctrine, is designed to resolve just this dilemma. Once we realize the role of the general concept in statements of number, we may appropriately identify what we have called the unit with the concept appearing in such statements:

> In the proposition "Jupiter has four moons", the unit is "moon of Jupiter". Under this concept falls moon I, and likewise also moon II, and moon III too, and finally moon IV. Thus we can say: the unit to which I relates is identical with the unit to which II relates, and so on. This

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[18] Frege, *FA*, p. 64.
[19] Frege, *FA*, p. 50.
[20] Frege, *FA*, p. 58.
[21] Ibid.

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gives us our identity. But when we assert the distinguishability of units, we mean that the things numbered are distinguishable.[22]

The identity (or unity) lies in the ability of different things to fall under the same *concept*; the difference (or plurality) lies in the fact that this subsumption does not make them into the same *object*. We can number distinct objects by bringing them together under a property they all share: but this does not mean that they have no *other* distinguishing properties, for then they would be the same object.

Now the very sort of dilemma outlined by Frege has its roots, according to Herbart, in a single perplexity — one which lies at the base of the major metaphysical quandaries.[23] From the fundamental Eleatic *aporia* of 'the one and the many' stem such difficulties as reconciling the one thing with many properties or the single ego with many presentations.[24] All such difficulties have something in common: they ascribe both unity and plurality to a single thing. In the case of number, the sole corrective, Herbart would counsel, is to keep in mind what he calls the "point of the relation," which here is "the concept of the counted."[25] If we keep our eyes fixed on this concept, Frege concurs, we may "easily solve the problem of reconciling the identity of units with their distinguishability."[26]

This close affinity between the views of Frege and Herbart, while hardly manifest to modern English readers, did not go unnoticed in Frege's own day. Reinhold Hoppe (1816-1900) in his review of the *Grundlagen* adverted to it several times, and so did Edmund Husserl (1859-1938) in his early work *Philosophie der Arithmetik*. If we briefly examine some quotations from these sources, then the cognizance of Frege's contemporaries will surely become clear.

Hoppe, as a proponent of psychologism, takes pains to point out some of Frege's chief errors. One of these, not surprisingly, involves Frege's approach to the concept: Frege, according to Hoppe, fails to take into proper account the "developmental history" of concepts. Hoppe chastizes Frege for his allegiance to the enemy camp, for being a logician for whom a concept is "not an achievement but rather an item lighted upon, which he takes into his work as a chemist would a fossil or a meteor-

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[22] Frege, *FA*, pp. 66-67.

[23] On this point see George H. Langley, "The Metaphysical Method of Herbart," *Mind*, n.s. 22 (1913): 68-69.

[24] Herbart, *SW*, vol. 4, p. 146.

[25] See Herbart, *SW*, vol. 6, p. 111 and vol. 8, pp. 134 and 182. For Herbart, the relation between the pure or undivided number and the general concept describes the "chief concept of arithmetic." Such "chief concepts" in every science, according to him, ought to be isolated for analysis by philosophy.

[26] Frege, *FA*, p. 66.

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ite."[27] In this very rebuke, Hoppe also has Herbart in range, for he immediately cites as "characteristic in this regard" Frege's direct appeal to Herbart on page iii on the *Grundlagen*.[28]

Very striking in this capacity, however, are Husserl's comments in *Philosophie der Arithmetik*. In the chapter devoted to "The Sense of the Statement of Number," Husserl first considers the view of John Stuart Mill (1806-1873) according to which numbers are "names of things." The chief alternative to this view is to be found, Husserl suggests, in the work of Frege and Herbart and he explicitly links the two thinkers:

> Entirely differently [from Mill] does Herbart judge it. "Each number," he says, "now relates . . . to a general concept of the counted" . . . And this view has in recent times found a zealous advocate in Frege. "The statement of number," he says briefly and clearly, contains "an assertion about a concept."[29]

So sure, in fact, is Husserl of the intimate connection between the two men on this very topic that he prefaces a discussion of Frege's views with the following remarks:

> The Herbartian view, according to which the statement of number relates to a concept, requires and merits a more precise investigation. Herbart himself has not carried out the foundation for this view with precision; we will therefore confine ourselves to the arguments of Frege, who applies all his care to this: to better develop Herbart's view and to demonstrate a positive as well as a critical path.[30]

Husserl thus definitively characterizes Frege as positively developing Herbart's insights into the structure of the statement of number. And such a distinct attribution is all the more remarkable when we recall that Frege reviewed *Philosophie der Arithmetik* for a philosophical journal.[31] So if this connection were far-fetched or unpalatable to him, would he not have

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[27] Reinhold Hoppe, Review of Frege's *Grundlagen, Archiv der Mathematik und Physik*, 2nd ser., pt. II, sect. 3 (1885): 29.

[28] This brings to light another possibly significant parallel between Frege and Herbart: both held to an anti-psychologism that proceeds via a sharp distinction between subjective ideas and objective concepts. Although Frege never gives any indication that his distinction is derived from an external source, there is some possibility that its origin lies in Herbart's work; Angelelli reports that: "A probable source is Eucken, where Herbart's distinction *Vorstellung-Begriff* is mentioned." See Ignacio Angelelli, *Studies on Gottlob Frege and Traditional Philosophy* (Dordrecht: D. Reidel, 1967), p. 86, n. 152. Rudolf Eucken (1846-1926) was a colleague of Frege's at Jena and Frege was certainly familiar with Eucken's book for he refers to it in the *Grundlagen*. In other writings, which Frege may or may not have known, Herbart clearly draws the distinction.

[29] Edmund Husserl, *Philosophie der Arithmetik*, ed. Lothar Eley (The Hague: Martinus Nijhoff, 1970), p. 161.

[30] Husserl, *PA*, p. 163.

[31] Frege, *CP*, pp. 195-209.

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taken that opportunity to set the record straight? If he had any desire to repudiate such a connection, would he not have ridiculed it in his lengthy and caustic review?[32]

Of course, in this last quotation, Husserl himself notes Herbart's critical weakness: the lack of any *systematic* development of his own insight. This lacking was corrected by Frege and his *Begriffsschrift*: for the idea that a statement of number makes an assertion about (or a predication of) a concept can only receive its full justification in a logical system that, among other things, allows for such first-order concepts to "stand to" these predications "in a relation similar to the one in which objects stand to" those first-order concepts.[33] That is to say, we need the notion of a *second-order* concept.[34]

The lack of a larger significance to Herbart's insight, absent the logical system of the *Begriffsschrift*, raises a sceptical question: should we not say at most that Frege was merely *aware* that Herbart had anticipated him? Here we should note that Frege certainly was aware that others had indeed anticipated the central doctrine, for he duly mentions Spinoza's *Epistolae*[35] and even Plato's *Hippias Major*.[36] How then can we characterize the reference to Herbart as different in kind from these?

The difference lies only in the hypothesis that Herbartianism may provide part of the significant and immediate historical *context* to Frege's work and hence not merely an ephemeral connection.[37] The direct refer-

[32] There are also other works from Frege's own day which allude to the Frege-Herbart connection, among them: G. F. Lipps, "Untersuchungen über die Grundlagen der Mathematik," *Philosophische Studien* 10 (1894): 200.

[33] Frege, *CP*, p. 337n.

[34] It must be noted, however, that Frege considers neither of these two "relations" ("falling *under*" and "falling *within*") to constitute a "third" thing. See Gottlob Frege, *Posthumous Writings*, ed. Hans Hermes *et al.* and trans. Peter Long *et al.* (Chicago: University of Chicago Press, 1979), p. 178.

[35] Frege, *FA*, p. 62.

[36] Frege, *PW*, p. 253.

[37] Such a context could help interpret, for example, the combative juxtapostion of Herbart and Kuno Fischer (1824-1907) on page iii of the *Grundlagen* as more than mere circumstance. Eventually, we might discover Herbart to be "a common *Urgroßvater*" of many different nineteenth-century philosophical and mathematical trends. See Barry Smith, "Wittgenstein and the Background of Austrian Philosophy," *Wittgenstein and his Impact on Contemporary Thought* (Vienna: Hölder-Pichler-Tempsky, 1978), p. 32. Smith correctly notes that "there was a time when Herbart's thought had come to achieve a position in (Catholic) Austria comparable to the State Hegelianism which held sway in Germany" (p. 31). But the oft-repeated perception of a dominant Hegelianism in Germany proper may not be adequate to the facts of *Frege's* own day. In 1877, for example, Wilhelm Wundt (1832-1920) offered a *précis* of the state of philosophy in Germany. While noting the continued influence of the schools of Kant, Hegel, and Herbart, he

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ence, which has been employed here, provides the link between text and context. But absent a complete survey of the relevant context, such a linkage cannot be fully evaluated. The delineation of Frege's intellectual milieu surely lies outside the scope of this single paper. That milieu is by necessity only indicated. But the context cannot be eliminated: to understand Frege we must know not only what he said but also the context in which he said it.[38]

found that the Hegelian school had "at the present day the fewest thoroughgoing adherents," counting only four major followers (including Fischer) in philosophy. In contrast, Wundt lists seven prominent disciples of Herbart. See Wilhelm Wundt, "Philosophy in Germany," *Mind* 2 (1877): 510-13.

[38] Special thanks to Michael Friedman and Penelope Maddy for their generous comments and encouragement on earlier drafts of this work. Its present form was shaped by the criticism of an anonymous referee for this Journal.

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